Automating resistivity measure using the four point probe method

César Besançon
Thomas Dias Botelho
Margot Lauret

SUMMARY

The project took place in the FEMTO-ST laboratory in Montbéliard, within the MINAMAS team, which specialize in micro- and nanoscale materials and surfaces. The team develops thin film materials for various applications, particularly in photovoltaics, where accurately measuring resistivity is crucial for optimizing performance.

Problematic:

The lab’s existing method for measuring resistivity of thin films relied on a system that required extensive manual handling, significant time, and did not account for the pressure applied by the probes. These shortcomings led to imperfect and inconsistent results.

Proposed solution:

To address these issues, a new setup was implemented:

- A Keithley 2400 SourceMeter, controlled by a python script, was introduced to automate electrical measurements.
- A MonoDAQ U-X module was used along DewesoftX to measure and regulate the pressure applied by the probes, ensuring consistent pressure across all samples.

This approach on the four-probe method (4-probe method), which isolate the resistance of the sample itself by eliminating the influence of cables and contacts resistance. In this method, the outer two probes generate a known current through the sample, while the inner two probes measure the voltage drop. Since no current flows through the inner probes, their resistance does not affect the measurement. Additionally, precise control of probe pressure improve contact quality between the probes and the samples, significantly reducing variability in results.

Project team:

This project was conduced by three second-year students in the Bachelor of Science In Technology (BUT) in Physical measurements at the IUT (University Institute of Technology) in Montbéliard. This program has provided us with skills in instrumentation, metrology and material analysis, essential for designing, and implementing innovative solution in the field of physical measurements.

Results:

The project resulted in a program capable of measuring thin film resistivity, with enhanced accuracy, and recording data in a format ready for analysis. This solution optimizes the measurement process while ensuring high reproducibility of results.

SET-UP

Analysis:

The thin films are fragile, the power flowing through them should be as low as possible to avoid any damage. Because of this, it is necessary to product a small and precise current in the µA range and to sense small voltages in the µV range.

Solution:

The first approach was to only use the MonoDAQ, but it is incapable of accurately producing such low currents and no solution was found to automatically process the results in DewesoftX.

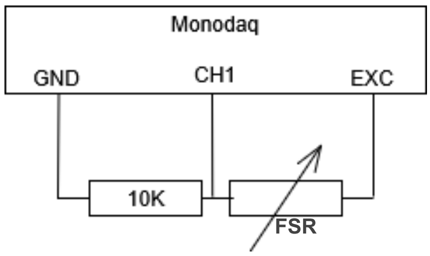
It was decided to use a Keithley SourceMeter to generate the currents and sense the voltages, the MonoDAQ to monitor the pressure and a python script to process and save the data and interface with the user.

A force sensing resistor (FSR) was used to measure the pressure applied, but this type of sensor is not well suited for this application. Changing to another technology would only require to change the DewesoftX setup file, so it was decided to use this sensor and simulate the pressure.

The force sensor used

Implementation:

The first step was the reading of the pressure sensor with the MonoDAQ.
The pressure sensor used is a force sensing resistor, whose resistance decreases when the pressure increases To measure the resistance, we use a simple voltage divisor. The DewesoftX setup file permit the easy processing of the sensor's output outside the python script. As the sensor is temporary, the output is only the measured resistance of the sensor.



Wiring of the pressure sensor

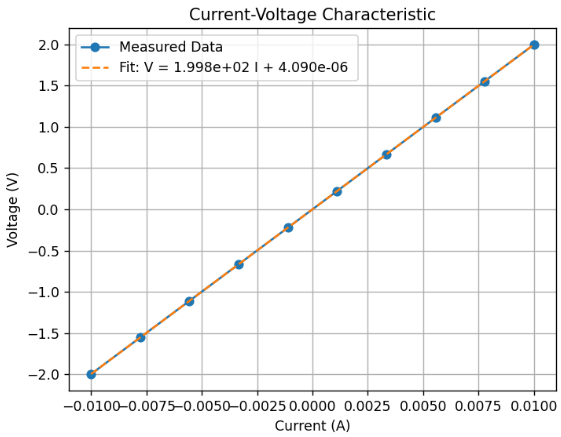
The second step was to use DCOM (Distributed Component Object Model) to retrieve the sensor data from DewesoftX to the python program, and the use of the pyVISA library to communicate with the Keithley using SCPI (Standard Commands for Programmable Instruments). This permits the full control of the instrument and enable the program to perform a lot of precise measure in a short amount of time.

The next part was to process and save the data. When plotting the voltage against the current, the slope is the resistance of the sample.

This can be verified by measuring known resistances. The resulting graph should be a straight line passing through the origin.

After that, the resistivity can be calculated with the formula: $\rho = 4.532 \times \frac{V}{I} \times t$

Where *4.532* is a factor associated with the shape of the probe array and *t* is the thickness of the thin film in cm. By comparing the results with those found using the old method, the system's accuracy can be confirmed.



Measure of a 200 ohms resistor

Every measurement is stored in an individual file, containing the measurements parameters and the results. At the end of the program, those files are combined into a single one.

The last part was to integrate every function into one single python script in the form of a state machine that allows the easy modification of the features without having to completely rewriting our code each time.

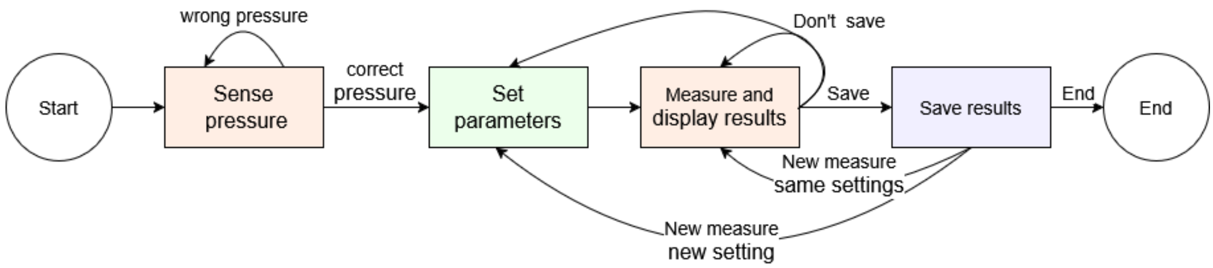
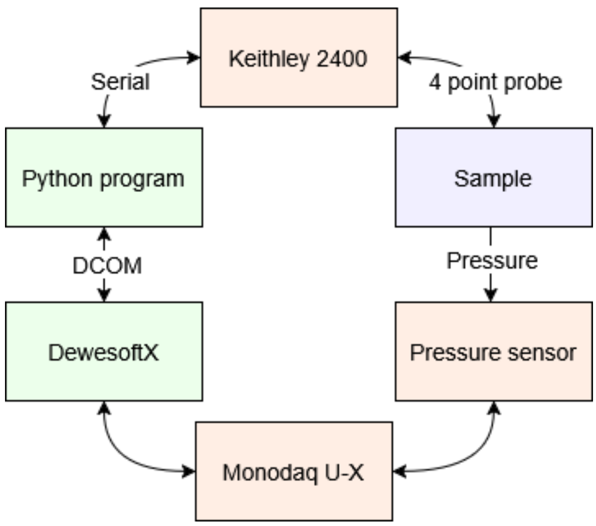


Diagram of the state machine



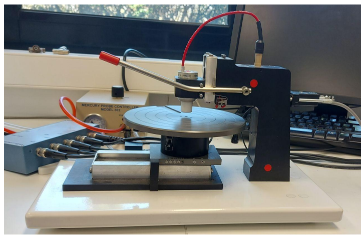
Block diagram of the project

List of necessary equipment:

As of now, the equipment required are:

- MonoDAQ-U-X
- Jandel multiposition wafer probe stand,
- Jandel cylindrical four points probe head,
- Keithley 2400 SourceMeter,
- CP 151 FSR pressure sensor,
- Computer equipped with dewesoftX and python,
- RS232 to USB adapter.

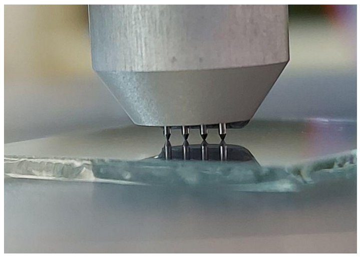
Monodaq U-X



Multiposition wafer probe stand

Keithley 2400 Sourcemeter

MEASUREMENTS


The probes in contact with a sample

To conduct a measurement, the user must start the Python program by clicking on the “main.py” file. A command line interface appears and a short amount of time is needed in order to set up the communication between the program and the MonoDAQ.
When everything is ready to go, the pressure measurements start and the user can set up the sample. When the pressure is comprised inside the set interval, the program will ask for the parameters of the measure:

- the name of the file where everything will be saved,
- the name of the sample tested,
- the thickness of the sample (in µm),
- the maximum current that will traverse the sample,
- the number of measure to perform.

The user is asked if the values entered are corrects, if they are, the measurement is performed. After a couple of seconds, the results are displayed as a graph and in the command line interface.

After a measurement, there is the possibility to save the results and to perform another measure using the same parameters or new ones.

At the end of the program, a single file containing all measured values and results is created in the same directory as the python program.

RESULTS & CONCLUSION

The team managed to develop a functional program, able to perform quick and precise measurements and save the results in a practical format.

The measure of the pressure is not yet functional, but it would be very easy to implement a suitable sensor (a load cell for example), to do so, only the dewesoftX setup file would have to be changed.

The program could be improved by adding a user interface to make it easier to operate by peoples not comfortable with a command line interface.

To make this project truly usable, it would be necessary to test it thoroughly to ensure its reliability and accuracy. We would also have to test every error possible and take them into account in order to improve the resiliency of the program.

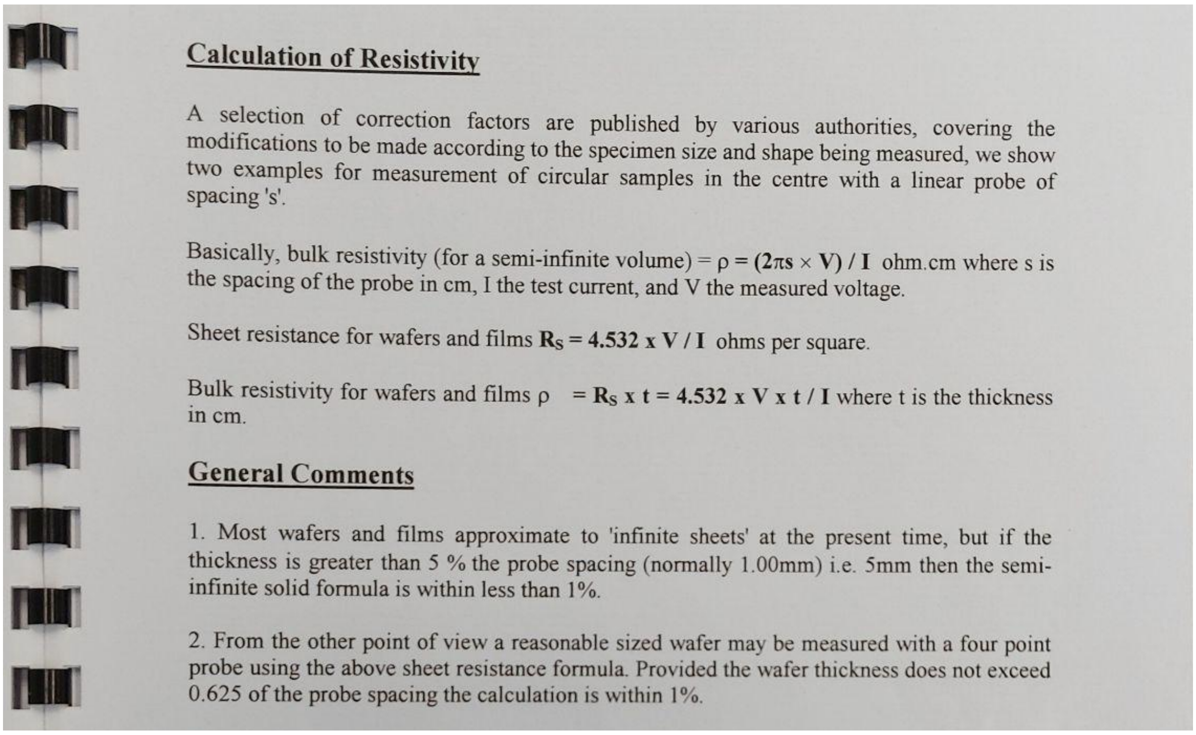
To improve upon this works, the MonoDAQ-U-X could be replaced with a more advanced system (from Dewesofts SIRIUS® line), resulting in extended capabilities and better performances.

Overall, this project was a success, although the pressure sensor is not yet useful, it could have an impact by greatly reducing the time and effort needed to measure resistivity and permitting to process the data more easily. We managed to improve our skills, particularly in programming, and to persevere through the many challenges we faced.

REFERENCES

An overview of various methods used to measure resistivity: [Thin Film Surface Resistivity]

Link to download the projects files: [4probes-controler (github)]



Calculation of Resistivity

A selection of correction factors are published by various authorities, covering the modifications to be made according to the specimen size and shape being measured, we show two examples for measurement of circular samples in the centre with a linear probe of spacing 's'.

Basically, bulk resistivity (for a semi-infinite volume) = $\rho = (2\pi s \times V) / I$ ohm.cm where s is the spacing of the probe in cm, I the test current, and V the measured voltage.

Sheet resistance for wafers and films $R_S = 4.532 \times V / I$ ohms per square.

Bulk resistivity for wafers and films $\rho = R_S \times t = 4.532 \times V \times t / I$ where t is the thickness in cm.

General Comments

1. Most wafers and films approximate to 'infinite sheets' at the present time, but if the thickness is greater than 5 % the probe spacing (normally 1.00mm) i.e. 5mm then the semi-infinite solid formula is within less than 1%.

2. From the other point of view a reasonable sized wafer may be measured with a four point probe using the above sheet resistance formula. Provided the wafer thickness does not exceed 0.625 of the probe spacing the calculation is within 1%.

Calculation of resistivity from the Jandel multiposition wafer probe manual